

RUNBAO (FRANK) DU

1005 E 60th St., Chicago, IL 60637 | runbao@uchicago.edu | +1 323-449-5920

EDUCATION

The University of Chicago, Chicago, IL

Bachelor of Science in Computational and Applied Mathematics, Minor in Astronomy and Astrophysics, May 2027

HONORS

Dean's List, Duolingo English Test Merit Award (2024)

EXPERIENCE

Lyra, Chicago, IL

Founder, April 2026 – Present

- Built and deployed a full-stack AI music platform where users upload a song and chat with an LLM that "listens" through a custom Python audio-analysis pipeline and generates on-the-fly visual scenes via Claude tool-use orchestrating Google's Gemini image model.
- Shipped it to production on Render with a Lovable-hosted frontend

Directed Research, Chicago, IL

Undergraduate Researcher, Feb 2026 – Present

- Analyzing artifacts in AI-generated music through Python pipelines in scale, including Crest Mean, MFCC Delta Mean Abs, Chroma Entropy Mean, Centroid-to-Rolloff Ratio etc.
- Building a deepfake detector that targets Suno-generated music and exploring prompts-engineering as a method of optimization

SUPERGroup Lab, Chicago, IL

Research Assistant, Nov 2025 – Present

- Examining behaviors of 8 major LLMs in value-based reasoning through persona and scenario descriptions
- Practicing prompts-engineering and output-formatting while also parsing and analyzing text data in bulk

UChicago Directed Reading Program, Chicago, IL

Mentee, Oct 2025 – Jan 2026

- Studied elliptic curves and algebraic geometry with a focus on cryptography; covered important topics like Affine and Projective Varieties, The Group Law of Elliptic Curves, and the Elliptic Curve Primality Test.

LEADERSHIP & ACTIVITIES

Southern California Math Competition, Los Angeles, CA

Vice President, Feb 2024 – May 2025

- Co-led the only student-initiated math competition that's recognized by the university; secured \$1000 + grants and held competitions per semester with 150+ participants coming from nearby high schools.
- Past participants have also won other awards, including IMO gold medals.

HIGHLEVEL COURSES

MATH: Theory of Numbers, Probability Theory, Seminar in Problem Solving, Abstract Linear Algebra, Analysis in $\mathbb{R}^n$ I-II-III

CS: Discrete Methods in Computer Science, Mathematical Foundations of Machine Learning, Standard Cryptography I (Online)

SKILLS

Languages: Chinese (Mandarin), English

Computer: Python, C++, Adobe Photoshop, Affinity Photo, LaTeX

AI: Claude Code, Claude Cowork, Cursor, Lovable, Suno, ACE Studio, Krea

Interests: Music Production (10+ Million Streams), Poetry, Boulderling, Ultimate Frisbee